

LESIKA DATA ANALYTICS TECHNICAL REPORT

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Module: CMPG325

Client: Lesika Data Analytics(Potchefstroom)

Client ID: CLI-144

Addressing Block: 192.168.60.0/24

Technical Challenge: Networking Troubleshooting: Fault Isolation.

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1. Introduction

This technical report covers the final network design, implementation, and testing for Lesika Data Analytics. Lesika Data Analytics is a small data analysis company situated in Potchefstroom. This project is a network design project for the module 325 (Computer Networks) and follows the client brief CLI-144.

The report in all will cover the following:

- The business background and client requirements.
- The physical and logical network topology
- The IP addressing and VLAN plan.
- Implementation for routers, switches, wireless, servers, and end devices.
- The assigned networking challenge: troubleshooting and fault isolation.
- The change request CR15, which requires a second internet connection for resilience.
- Testing and evidence
- Design limitation and future improvements.

The network was developed, implemented, as well as simulated in Cisco Packet Tracer. Find all the devices configurations, testing evidence and other documentation on the following GitHub repository:

2. Client background and Requirements

2.1 Client Background

Lesika Data Analytics is a small data analysis company based/situated in Potchefstroom. The company's services include data cleaning and analysis, as well as building dashboards and producing analysis reports. Lesika Data Analytics is a small company; it operates as a SOHO (small office/home office) environment with 20 employees, with expected growth to about 25 over the next 2 years.

The company needs a secure and scalable network that supports the following:

- Wireless and wired connectivity for staff.
- Shared access to file, database, and print services.
- Internet access.
- Separate guest Wi-fi with isolation from internal company resources.
- Resilience with backup internet.

2.2 Employee Headcount

Role	Number
Director	1
Finance Manager	1
Admin/HR	1
Data Analysts	8
Data Scientist / Engineers	4
IT Support	2
Sales	3
Total	20

2.3 Client requirements

1. A network that currently supports 20 employees and can scale to 25.
2. Wired and Wireless network access.
3. Segregation of staff, guest, server, printer, and management traffic using VLANs.
4. Provide DHCP and DNS services automatically.
5. Provide shared file and database services.
6. Provide secure internet access using NAT.
7. Protect company data from guests.
8. Support devices such as VoIP phones and CCTV devices for future additions
9. A Network that is easy to troubleshoot and document.
10. Use Cisco Packet Tracer for implementation.
11. Be manageable on an SOHO budget.
12. Integrate a second internet connection for CR15.
13. Reserve IP addressing for a future office branch that is to be opened in 18 months.
14. Demonstrate network troubleshooting(fault isolation)

3. Network Design

3.1 Design Overview

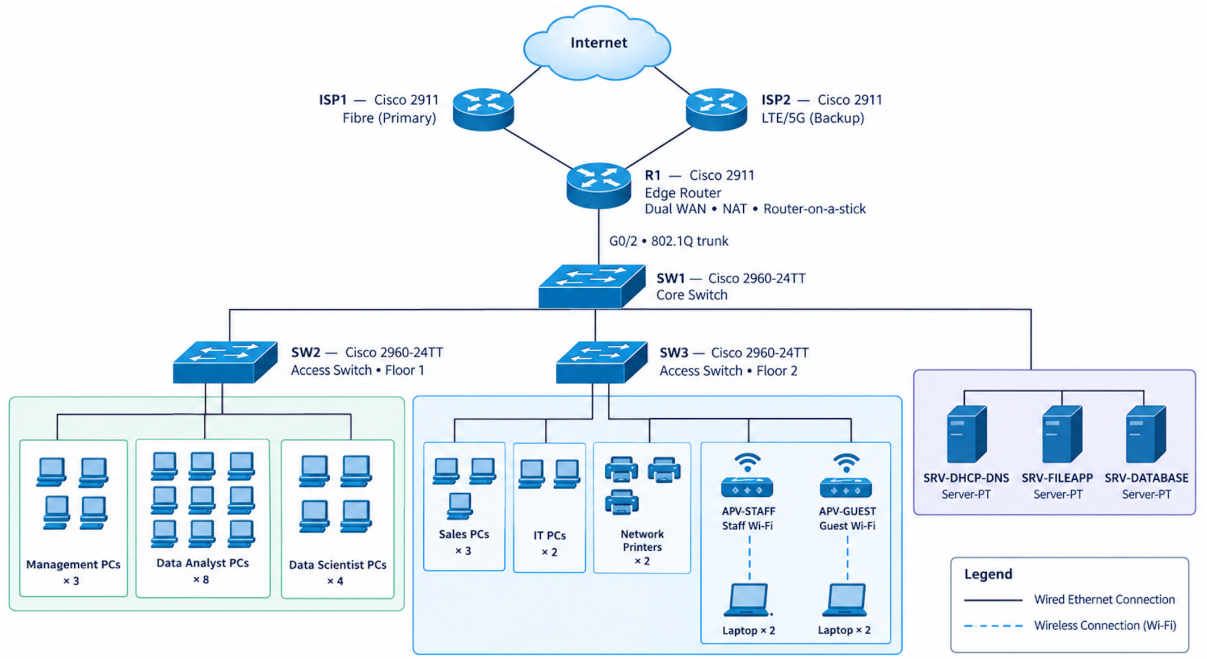
The network is a simple three-layer design that features the following:

- Edge layer: Two ISP routers(simulated) and one 2911 router (R1)
- Core layer: A Cisco 2960-24TT switch, SW1, that acts as the core and is trunked to two access switches.
- Access layer: Two Cisco 2960-24TT Switches, SW2 and SW3. They provide access ports to wired devices. Two wireless access points also provide coverage: one for guests and one for staff.

Servers(for services), PCs, printers, and wireless access points all connect to the core and access switches using the correct VLANs

3.2 Physical Topology

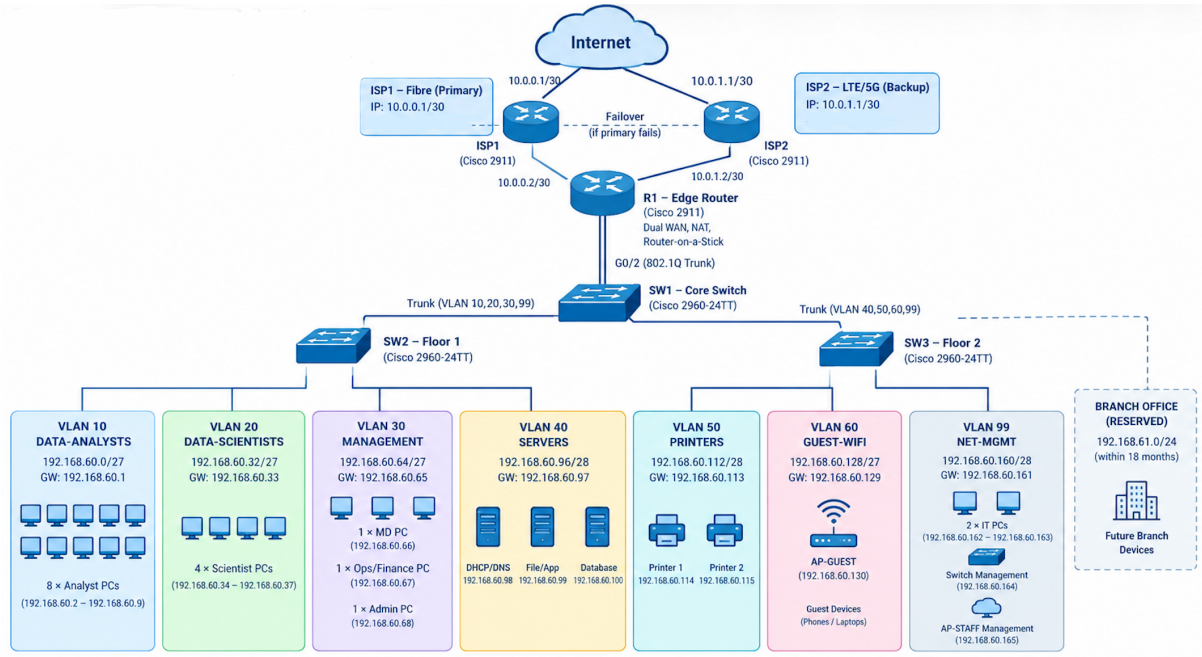
The physical topology is:



Device	Model	Role
ISP1	Cisco 2911	Simulated Fibre ISP-main
ISP2	Cisco 2911	Simulated LTE/5G ISP-Backup
R1	Cisco 2911	Edge router, dual WAN, NAT, router-on-a-stick
SW1	Cisco 2960-24TTT	Core switch
SW2	Cisco 2960-24TTT	Access point for management, analysts and data scientists
SW3	Cisco 2960-24TTT	Access point for Sales, IT, printers and APs
SRV-DHCP-DNS	Server-PT	DHCP and DNS services
SRV-FILEAPP	Server-PT	Application and files services
SRV-DATABASE	Server-PT	Database server
AP-STAFF	AccessPoint-PT	Staff Wi-Fi
AP-GUEST	AccessPoint-PT	Guest Wi-Fi

3.3 Logical Topology

The logical topology uses VLANs to segment the network and 802.1Q trunking to carry VLANs between switches.



VLAN	Name	Purpose	Subnet	Gateway
10	DATA ANALYTS	Data analyst PCs	192.168.60.0/27	192.168.60.1
20	DATA ENGINNERS	Data engineers PCs	192.168.60.32/27	192.168.60.33
30	MANAGEMENT	Management and sales PCs	192.168.60.64/27	192.168.60.64
40	SERVERS	File, DHCP, DNS, DB	192.168.60.96/27	192.168.60.97
50	PRINTERS	Network printers	192.168.60.112/27	192.168.60.113
60	GUEST-WIFI	Guest wireless	192.168.60.128/27	192.168.60.129
99	NET-MGMT	Switches, IT PCs and APs	192.168.60.160/27	192.168.60.161

3.4 Device Selection Justification

Device	Reason for selection
Cisco 2911 Router	A reliable edge router equipped with multiple Gigabit Ethernet ports, supporting dual WAN, NAT and ACLs
Cisco 2960-24TT Switches	Standard access and core switches featuring IEEE 802.1Q Trunk support and VLAN capabilities
Server-PT devices	Simulation of DHCP, DNS, file and database services.
AccessPoint-PT	Provision of wireless SSID management for guests and staff networks
PC-PT and Laptop-PT	Representation of employee workstations and wireless client devices
Printer-PT	Simulation of shared network printers within the office environment.

4. IP Addressing and VLAN Plan

4.1 Assigned Addressing Block

The assigned block is 192.168.60.0/24, which gives 256 addresses, 254 usable.

4.2 Subnetting

Variable Length Subnet Masking (VLSM) was used to subnet the networks based on the current and expected host requirements

VLAN	Hosts needed	Chosen size	Usable
10	25	/27	30
20	25	/27	30
30	25	/27	30
40	8	/28	14
50	4	/28	14
60	25	/27	30

99	8	/28	14

4.3 Full IP addressing Table

VLAN	Subnet	Network	Usable range	Broadcast	Gateway
10	/27	192.1268.60.0	.1-.30	.31	.1
20	/27	192.1268.60.32	.33 -.62	.63	.33
30	/27	192.1268.60.64	.65 - .94	.95	.65
40	/28	192.1268.60.96	.97- .110	.111	.97
50	/28	192.1268.60.112	.113 - .126	.127	.113
60	/27	192.1268.60.128	.129 - .158	.159	.129
99	/28	192.1268.60.160	.161 - 174	.175	161

4.4 WAN Addressing

Link	Network	R1 side	ISP side
R1-ISP1	10.0.0.0/30	10.0.0.2	10.0.0.1
R1-ISP2	10.0.1.0/30	10.0.1.2	10.0.1.1
ISP1 public	203.0.113.0/30	-	203.0.113.1
ISP2 public	198.51.100.0/30	-	198.51.100.1

4.5 Static IP Assignments

Device	**IP Address**
R1 G0/2.10	192.168.60.1
R1 G0/2.20	192.168.60.33
R1 G0/2.30	192.168.60.65
R1 G0/2.40	192.168.60.97
R1 G0/2.50	192.168.60.113
R1 G0/2.60	192.168.60.129
R1 G0/2.99	192.168.60.161
SRV-DHCP-DNS	192.168.60.98
SRV-FILEAPP	192.168.60.99
SRV-DATABASE	192.168.60.100
PRINTER-01	192.168.60.114
PRINTER-02	192.168.60.115
PC-IT1	192.168.60.162
PC-IT2	192.168.60.163

4.6 DHCP Ranges

VLAN	DHCP range
10	.2 -.30
20	.34 -.62
30	.66 -.94
60	.130 -.158

4.7 Growth and Branch Office Reservation

The following space is reserved to support growth and a future office branch that is planned to happen in 18 months:

- 192.168.60/28 – Main Office Growth
- 192.168.60/26 - Future office branch
- 192.168.60/30 – Main Branch WAN link
- 192.168.60/30 – Reserved

5. Implementation Summary

5.1 VLAN Configuration

All Switches(SW1, SW2 and SW3) were set up with the following VLANS: 10, 20, 30, 40,50, 60, and 99. The access ports were matched to the correct VLANs according to the physical topology. Trunk links between SW1 and SW2, also between SW1 and SW3 used 802.1Q encapsulation and allowed the necessary VLANs.

5.2 Router-on-a-stick

R1 was set up with sub-interfaces on G0/2, one for each VLAN, using 802.1Q encapsulation. Each sub-interface got the VLAN’s gateway IP and acted as the default gateway for the devices in that VLAN.

5.3 DHCP and DNS

The DHCP and DNS server was set up/ configured as follows:

- Four DHCP pools were established for VLANs 10, 20, 30 and 60.
- A range of DNS A-records were created to support internal name resolution.

R1 router was set up with the ip helper-address 192.168.60.98 on each sub-interface to forward DHCP requests from client VLANs to the server.

5.4 Wireless

Two access point were set up:

- AP-STAFF with SSID: LDA-Staff. WPA2-PSK
- AP-GUEST with SSID: LDA-Guest. WPA2-PSK

AP-STAFF connects to VLAN 10(guest) , and AP-GUEST connects to VLAN60(guest). Laptops were setup with a wireless network cards and connected to the correct Wi-Fi networks.

5.5 NAT and Guest Isolation

NAT overload was set up on the R1 router to change the internal IP addresses to the WAN interface. An ACL9 ACL 100 was applied inbound on the guest VLAN sub-interface to block guest traffic to the internal network while allowing internet access.

5.6 CR15- Dual WAN Resilience

The R1 router was set up with:

- A main fixed route through ISP1: ip route 0.0.0.0.0.0.0.10.0.0.1
- A backup fixed route with priority 10 ip route 0.0.0.0.0.0.0.10.0.1.1 10

When the main connection is working, traffic goes through ISP1. If ISP1 stops working, the backup route will start working, and traffic will automatically switch to ISP2. When ISP1 works again, traffic returns to the main route.

6. Assigned Networking Challenge: Fault Isolation

6.1 Overview

The assigned networking task is fixing network problems by finding where the issue is. Five real faults were introduced, found, fixed, and checked. Each fault affects a different layer of the OSI model, showing a step-by-step troubleshooting approach.

6.2 Fault 1: Disabled Trunk port between SW1 and SW2

Symptom: PC1-AN1 cannot reach its gateway; the link light between SW1 and SW2 is red.

Diagnostic: The command “show interfaces fa0/2” on SW1 shows is it administratively down. The command show interfaces fa0/1 on SW2 shows it is down. Comparing the two sows the problem is on SW.

Fix: Turn on the interface fa0/2 on SW1 using the *no shutdown* command.

Verification: Link light turns green.

6.3 Fault 2: Missing VLAN on Trunk

Symptom: PC-DS1 cannot reach its VLAN 20 gateway 192.168.60.33 or the file server 192.168.60.98

Diagnostic: The command “show interfaces trunk” on SW1 shows that Fa0/2 allows VLAN 10, 30, 40, 50, 60, and 99, but not VLAN 20.

Fix: Allow VLAN 20 on the trunk port Fa0/2 on SW1 by using the command ‘switchport trunk allowed vlan 10, 20, 30, 40, 50, 60,99”

Verification: PC-AN1 can ping 192.168.60.33 and 192.168.60.98 successfully.

6.4 Fault 3: Wrong IP address on a PC

Symptom: PC-AN3 cannot reach its gateway, the server or any VLANs

Diagnostic: Running ipconfig on PC-AN3 shows the IP address is 10.10.10.10 with mask 255.255.255.224, gateway 192.168.60.1 and DNS 192.168.60.98.

Fix: Set the correct IP settings on PC-AN3: 10 192.168.60.28, mask 255.255.255.224, gateway 192.168.60.1, and DNS 192.168.60.98.

Verification: PC-AN3 can now successfully ping 192.168.60.1 and 192.168.60.98.

6.5 Fault 4: Missing IP Helper-address on VLAN 20 Sub-interface

Symptom: PC-DS3 cannot obtain an IP address via DHCP ipconfig shows 0.0.0.0 or an APIPA address.

Diagnostic: The command “show ip interface g0/2.20’ on router R1 shows “Helper address is not set. The command “ show ip interface g0/2.10” shows “Helper address is 192.168.60.98’. Comparing the two shows and confirms the helper address is missing.

Fix: Add the helper address 192.168.60.34 on R1 interface g0/2.20.

Verification: PC-DS3 receives an IP in the range 192.168.60.34-.62, and with gateway 192.168.60.33.

6.6 Fault 5 : Missing DNS Record

Symptom: PC-AN1 can ping the file server by IP 192.168.60.99, but cannot resolve ‘fileserver.lesika.local’.

Diagnostic: Diagnostic: In SRV-DHCP-DNS under Services and DNS, the A record for fileserver.lesika.local is missing. Other records still work correctly.

Fix: Add back the A record for fileserver.lesika.local with the address 192.168.60.99.

Verification: nslookup fileserver.lesika.local resolves to 192.168.60.99 and pinging fileserver,lesika.local succeeds.

7. Change Request CR15: Dual WAN Resilience

CR15 required adding a second Internet connection for backup. This was done as follows:

- ISP1 is the primary connection, simulated by a Cisco 2911 router.
- ISP2 is the backup connection, simulated by another Cisco 2911 router.
- R1 is connected to both ISPs via G0/0 and G0/1.
- A primary default route points to ISP1.
- A backup static default route (AD 10) points to ISP2.
- When ISP1 fails, the floating route takes over automatically.
- When ISP1 recovers, the primary route is preferred again.

The backup switch was tested by turning off G0/0 on R1 and confirming that the main gateway changed from 10.0.0.1 to 10.0.1.1. Screenshots of both situations are included in the test evidence.

8. Testing and Verification

All tests were carried out in Cisco Packet Tracer. Evidence is provided as screenshots in the GitHub repository.

1	DHCP from wired VLAN 10 PC	PASS
2	Ping default gateway from VLAN 10	PASS
3	Cross-VLAN ping to servers	PASS
4	DNS resolution	PASS
5	Wireless staff connectivity	PASS
6	Wireless guest connectivity	PASS
7	CR15 failover – backup route active	PASS
8	CR15 failover – primary route restored	PASS
9	Guest isolation	PASS
10	NAT translations on R1	PASS
11	VLAN configuration on SW1	PASS
12	Trunk configuration on SW1	PASS
13	Router sub-interfaces on R1	PASS
14	Fault 1 – Disabled trunk port	PASS
15	Fault 2 – Missing VLAN on trunk	PASS
16	Fault 3 – Wrong IP address on PC	PASS
17	Fault 4 – Missing DHCP relay	PASS

18	Fault 5 – Missing DNS record	PASS
19	File reopens and reproduces working state	PASS

9. Future Improvements

9.1 Future Improvements

- Replace router-on-a-stick with a Layer 3 switch at the center to reduce the workload on R1.
- Add a separate firewall between R1 and the center for better security control.
- Provide a public DNS resolver for guests.
- Add backup switches by stacking or using two main switches.
- Complete the setup of VoIP and CCTV VLANs with priority settings for better performance.
- Implement a branch office as per the reserved addressing in 192.168.60.192/26.

10. Conclusion

The Lesika Data Analytics network was successfully planned, built, and tested in Cisco Packet Tracer. The network meets all client needs, including:

- VLAN-segmented connectivity for 20 employees
- DHCP and DNS services with cross-VLAN relay
- Wireless staff and guest networks with isolation
- Dual WAN resilience (CR15)
- NAT overload for Internet access
- Five documented fault isolation scenarios

The design can grow with needs, is well documented, and is easy to fix problems. It is ready to be used in a real small office or home office (SOHO).

11. References

1. Cisco Systems. (2024). *Cisco Packet Tracer User Guide*. Cisco Press.
2. Cisco Systems. (2023). *Interconnecting Cisco Networking Devices, Part 1 (ICND1)*. Cisco Press.
3. Odom, W. (2020). *Cisco CCNA 200-301 Official Cert Guide, Volume 1*. Cisco Press.
4. Odom, W. (2020). *Cisco CCNA 200-301 Official Cert Guide, Volume 2*. Cisco Press.
5. RFC 1918 – *Address Allocation for Private Internets*.
6. RFC 2131 – *Dynamic Host Configuration Protocol*.
7. RFC 1035 – *Domain Names – Implementation and Specification*.

12. Appendices

Appendix A: Device Configurations:

Full device configurations are available in the GitHub repository under 05-configurations/:

Appendix B: Screenshot Index

All screenshots are available in 06-testing-evidence/screenshots/ in the GitHub repository.

Appendix C: Fault Isolation Documentation

Full fault documentation is available in 07-troubleshooting/fault-isolation-scenarios.md.

Appendix D: GitHub Repository:

https://github.com/KatDEv98/Final_Year_CMPG325_Networking_Project_Lesika_Data_Analytics_Network

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End of Technical Report